MATH1211

Chapter 7: ODE

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ORDINARY DIFFERENTIAL EQUATION

A differential equation is an equation that relates a function to its own derivatives.

- **Order:** The highest derivative present in the equation (equals the number of arbitrary constants).
- **Degree:** The power of the highest-order derivative.

$$\frac{d^3y}{dx^3} + 4\left(\frac{dy}{dx}\right)^4 - 5y = 0$$

- Order → 3
- Degree → 1

Examples of Differential Equations

Example 1: $y^2 = 4ax$

Differentiating with respect to x:

$$2yy_1 = 4a \Longrightarrow a = \frac{yy_1}{2}$$

Substitute a back into $y^2 = 4ax$:

$$y^2 = 4\left(\frac{yy_1}{2}\right)x = 2xyy_1 \Longrightarrow y_1 = \frac{y}{2x}$$

Example 2: $y = a\cos x + b\sin x$

Differentiating with respect to x:

$$y_1 = -a\sin x + b\cos x$$

$$y_2 = -a\cos x - b\sin x = -(a\cos x + b\sin x)$$

$$y_2 = -y \Longrightarrow y_2 + y = 0$$

Example 3: $y = a\cos(\alpha x) + b\sin(\alpha x)$

$$y_1 = -\alpha a\sin(\alpha x) + \alpha b\cos(\alpha x)$$

$$y_2 = -\alpha^2 a\cos(\alpha x) - \alpha^2 b\sin(\alpha x) = -\alpha^2(a\cos(\alpha x) + b\sin(\alpha x))$$

$$y_2 = -\alpha^2 y \Longrightarrow y_2 + \alpha^2 y = 0$$

Example 4: $x^2 + y^2 + 2gx + 2fy + c = 0$

Differentiating with respect to x:

$$2x + 2yy_1 + 2g + 2fy_1 = 0$$

$$x + yy_1 + g + fy_1 = 0$$

Differentiating a second time:

$$1 + y_1^2 + yy_2 + fy_2 = 0$$

$$1 + y_1^2 + y_2(y + f) = 0 \quad \text{— (i)}$$

$$y + f = -\frac{1 + y_1^2}{y_2} \quad \text{— (ii)}$$

Differentiating equation (i) with respect to x:

$$2y_1y_2 + y_1y_2 + (y + f)y_3 = 0$$

$$3y_1y_2 + (y + f)y_3 = 0$$

Substitute $(y + f)$ from equation (ii):

$$3y_1y_2 + \left(-\frac{1 + y_1^2}{y_2}\right)y_3 = 0$$

$$(1 + y_1^2)y_3 - 3y_1y_2^2 = 0$$

Equations of the 1st Order and 1st Degree

A differential equation of first order and first degree can be written in the form:

$$M\,dx + N\,dy = 0$$

where both M and N are functions of x and y, or constants not involving derivatives. The general solution contains only one arbitrary constant.

Methods of Solution

1. Separation of Variables If the equation can be put in the form $f_1(x)dx + f_2(y)dy = 0$, then it can be solved by integrating:

$$\int f_1(x)dx + \int f_2(y)dy = C$$

- **Problem 1:** $(1 + y^2)dx + (1 + x^2)dy = 0$

  Dividing by $(1 + x^2)(1 + y^2)$:

  $$\frac{dx}{1 + x^2} + \frac{dy}{1 + y^2} = 0$$

  Integrating:

$$\int \frac{dx}{1+x^2} + \int \frac{dy}{1+y^2} = C \Longrightarrow \tan^{-1}x + \tan^{-1}y = C$$

- **Problem 2:** $x(y^2+1)dx + y(x^2+1)dy = 0$

  Dividing by $(x^2+1)(y^2+1)$:

$$\frac{x}{x^2+1}dx + \frac{y}{y^2+1}dy = 0$$

  Integrating:

$$\frac{1}{2}\ln(x^2+1) + \frac{1}{2}\ln(y^2+1) = C' \Longrightarrow (x^2+1)(y^2+1) = C$$

- **Problem 3:** $(x+y)^2\frac{dy}{dx} = a^2$

  Let $x+y = v \Longrightarrow 1 + \frac{dy}{dx} = \frac{dv}{dx} \Longrightarrow \frac{dy}{dx} = \frac{dv}{dx} - 1$

$$v^2\left(\frac{dv}{dx} - 1\right) = a^2$$

$$\frac{dv}{dx} = 1 + \frac{a^2}{v^2} = \frac{a^2+v^2}{v^2}$$

$$dx = \frac{v^2}{a^2+v^2}dv = \left(1 - \frac{a^2}{a^2+v^2}\right)dv$$

  Integrating:

$$\int dx = \int \left(1 - \frac{a^2}{a^2+v^2}\right)dv$$

$$x + C = v - a^2 \cdot \frac{1}{a}\tan^{-1}\left(\frac{v}{a}\right)$$

$$x + C = (x+y) - a\tan^{-1}\left(\frac{x+y}{a}\right) \Longrightarrow y + C' = a\tan^{-1}\left(\frac{x+y}{a}\right)$$

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2. Homogeneous Equations If M and N in $Mdx + Ndy = 0$ are homogeneous functions of x and y of the same degree, the equation can be written as:

$$\frac{dy}{dx} = f\left(\frac{y}{x}\right)$$

It can be solved by substituting $y = vx \Longrightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$.

- **Example:** Solve $(x^2+y^2)dx - 2xy\,dy = 0$

$$\frac{dy}{dx} = \frac{x^2 + y^2}{2xy}$$

Put $y = vx \Longrightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$:

$$v + x\frac{dv}{dx} = \frac{x^2 + v^2x^2}{2vx^2} = \frac{1 + v^2}{2v}$$

$$x\frac{dv}{dx} = \frac{1 + v^2}{2v} - v = \frac{1 - v^2}{2v}$$

$$\frac{dx}{x} - \frac{2v}{1 - v^2}dv = 0$$

Integrating:

$$\ln x + \ln(1 - v^2) = \ln C \Longrightarrow x(1 - v^2) = C$$

Substituting $v = \frac{y}{x}$:

$$x\left(1 - \frac{y^2}{x^2}\right) = C \Longrightarrow x^2 - y^2 = Cx$$

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3. Exact Equations The differential equation $Mdx + Ndy = 0$ is exact if there exists a function $u(x, y)$ such that $Mdx + Ndy = du$. The necessary and sufficient condition for exactness is:

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

- **Example:** Solve $(2x^3 + 4y)dx + (4x + y - 1)dy = 0$

  Here, $M = 2x^3 + 4y$ and $N = 4x + y - 1$.

  $$\frac{\partial M}{\partial y} = 4, \quad \frac{\partial N}{\partial x} = 4$$

  Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, the equation is exact.

  Solution Calculation:

  $$\int M\,dx = \int (2x^3 + 4y)\,dx = \frac{x^4}{2} + 4xy$$

  $$\int N\,dy = \int (4x + y - 1)\,dy = 4xy + \frac{y^2}{2} - y$$

Combining terms and avoiding repetitions gives the general solution:

$$\frac{x^4}{2} + 4xy + \frac{y^2}{2} - y = C$$

Linear Differential Equations (First Order)

- **Definition:** A linear equation is an algebraic equation in which each term has an exponent of 1 (or 0), and its graph forms a straight line when plotted on a coordinate plane.
- An equation of the form $\frac{dy}{dx} + Py = Q$, where P and Q are functions of x or constants, is called a linear equation of the first order.

- **Problem :** Solve $\frac{dy}{dx} + \frac{y}{x} = x^2$.
  - $P = \frac{1}{x}, Q = x^2$.
  - Integrating Factor (IF) $= e^{\int P\,dx}$
  - $= e^{\int \frac{1}{x} dx}$
  - $= e^{\ln x}$
  - $= x$.
  - Multiplying by IF: $d(yx) = x^3\,dx$.
  - Integrating: $yx = \int x^3\,dx = \frac{x^4}{4} + c$
  - $\Longrightarrow y = \frac{x^3}{4} + \frac{c}{x}$.

$$\int u\,dv = uv - \int v\,du$$

Step 1: Choose u and dv

- Let $u = y + 1 \implies du = dy$
- Let $dv = e^{-y}\,dy \implies v = -e^{-y}$

Step 2: Apply the integration by parts formula

$$\int (y+1)e^{-y}\,dy = (y+1)(-e^{-y}) - \int (-e^{-y})\,dy$$

$$, = -(y+1)e^{-y} + \int e^{-y}\,dy$$

- **Problem :** Solve $(x + y + 1)\frac{dy}{dx} = 1$.
  - $\frac{dx}{dy} = x + y + 1 \Longrightarrow \frac{dx}{dy} - x = y + 1$.
  - Here $P = -1, Q = y + 1$.
  - IF $= e^{\int -1\,dy} = e^{-y}$.
  - Solution: $xe^{-y} = \int (y+1)e^{-y}\,dy$.
  - Using integration by parts: $\int (y+1)e^{-y}\,dy = -(y+1)e^{-y} - \int -e^{-y}\,dy = -e^{-y}(y+2)$.
  - $xe^{-y} = -e^{-y}(y+2) + c \Longrightarrow x + y + 2 = ce^{y}$.

- **Problem :** Solve $\cos^2 x \frac{dy}{dx} + y = \tan x$.
  - Divide by $\cos^2 x$: $\frac{dy}{dx} + y\sec^2 x = \tan x \sec^2 x$.
  - IF $= e^{\int \sec^2 x\,dx} = e^{\tan x}$.

- Integration: $ye^{\tan x} = \int \tan x \sec^2 x e^{\tan x}\, dx.$
- Let $u = \tan x$, $du = \sec^2 x\, dx$:

$$\int ue^u\, du = ue^u - e^u = e^u(u-1)$$

- $ye^{\tan x} = e^{\tan x}(\tan x - 1) + c \Longrightarrow y = \tan x - 1 + ce^{-\tan x}.$

Bernoulis Equation (non linear)

- **Problem :** Solve $\frac{dy}{dx} + \frac{y}{x} = x^4 y^4 \cos x.$
  - Divide by y^4: $y^{-4}\frac{dy}{dx} + \frac{1}{x}y^{-3} = x^4 \cos x.$
  - Let $z = y^{-3} \Longrightarrow \frac{dz}{dx} = -3y^{-4}\frac{dy}{dx} \Longrightarrow y^{-4}\frac{dy}{dx} = -\frac{1}{3}\frac{dz}{dx}.$
  - Substitute: $-\frac{1}{3}\frac{dz}{dx} + \frac{z}{x} = x^4 \cos x \Longrightarrow \frac{dz}{dx} - \frac{3}{x}z = -3x^4 \cos x.$
  - IF $= e^{\int -\frac{3}{x}dx} = e^{-3\ln x} = x^{-3}.$
  - Multiply by IF: $zx^{-3} = \int -3x^4 \cos x \cdot x^{-3}\, dx = -3\int x \cos x\, dx.$
  - $\int x \cos x\, dx = x \sin x + \cos x.$
  - $y^{-3}x^{-3} = -3(x \sin x + \cos x) + c.$

2nd Order Homogeneous Linear Differential Equation

- 2nd order linear differential equation with constant co-efficient.
- If right hand member is zero, then it is homogeneous, or it is non homogeneous.

  P. $\neq 0$

  Problem :

  Solve $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 0 \quad \ldots(1)$

  Let $y = e^{mx}$ be a solution of above equation.

$$\therefore \frac{dy}{dx} = me^{mx}$$

$$\frac{d^2y}{dx^2} = m^2 e^{mx}$$

  Then $m^2e^{mx} + 3me^{mx} + 2e^{mx} = 0$

$$\Rightarrow e^{mx}(m^2 + 3m + 2) = 0$$

$$e^{mx} \neq 0 \quad \therefore m^2 + 3m + 2 = 0 \quad \Rightarrow m = -1, -2$$

$\therefore$ The general solution is $y = c_1 e^{-x} + c_2 e^{-2x}$

In a non homogeneous equation right hand member is a function of x. It can be i) Polynomial ii)Exponential iii) Trigonometric

Problem : Solve $(D^2 + 4)y = x^2$

$$D^2y + 4y = 0 \quad \text{let } y = e^{mx} \text{ be a solution}$$

$$y = e^{mx}, \quad Dy = me^{mx}, \quad D^2y = m^2e^{mx}$$

$$\therefore \Rightarrow m^2e^{mx} + 4e^{mx} = 0 \Rightarrow e^{mx}(m^2 + 4) = 0$$

$$m = \pm 2i$$

$$\therefore \underset{\text{complementary}}{y_c} = c_1 e^{2ix} + c_2 e^{-2ix}$$

$$= c_1(\cos 2x + i\sin 2x) + c_2(\cos 2x - i\sin 2x)$$

$$= (c_1 + c_2)(\cos 2x) + \sin 2x(c_1 i - c_2 i)$$

$$= A\cos(2x) + B\sin 2x$$

Now Particular integral (PI) $\Rightarrow \frac{1}{D^2+4}x^2$

$$= \frac{1}{4\left(1 + \frac{D^2}{4}\right)}x^2 = \frac{1}{4}\left(1 + \frac{1}{4}D^2\right)^{-1}x^2$$

$$= \frac{1}{4}\left(1 - \frac{1}{4}D^2 + \frac{1}{16}D^4 - \cdots\right)x^2$$

1. **Calculate the Derivatives:**
   - $D(x^2) = 2x$
   - $D^2(x^2) = 2$
   - $D^4(x^2) = 0$
2. **Evaluate the Particular Integral (**PI**):**

$$\text{PI} = \frac{1}{4}\left(x^2 - \frac{1}{4}D^2(x^2)\right)$$

$$\text{PI} = \frac{1}{4}\left(x^2 - \frac{1}{4}(2)\right)$$

$$\text{PI} = \frac{1}{4}\left(x^2 - \frac{1}{2}\right) = \frac{x^2}{4} - \frac{1}{8}$$

(This explains the handwritten $\left(x^2 - \frac{1}{2}\right)$ term where the exponent 2 was omitted on x in the handwritten notes).

3. **General Solution (**GS**):**

$$y = y_c + y_p$$

$$y = A\cos 2x + B\sin 2x + \frac{1}{4}\left(x^2 - \frac{1}{2}\right)$$

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Example : Solve $(D-3)^2 y = 2e^{4x}$

Find the Complementary Function (C.F.**):**

Set the homogeneous part to zero: $(D-3)^2 y = 0$. The auxiliary equation is:

$$(m-3)^2 = 0 \Longrightarrow m_1 = 3,\ m_2 = 3$$

Since the roots are real and repeated:

$$\text{C.F.} = (A + Bx)e^{3x}$$

Find the Particular Integral (P.I.**):**

$$\text{P.I.} = \frac{1}{(D-3)^2}(2e^{4x})$$

Using the exponential rule $\frac{1}{f(D)}e^{ax} = \frac{1}{f(a)}e^{ax}$ where $a = 4$:

$$\text{P.I.} = \frac{2e^{4x}}{(4-3)^2} = \frac{2e^{4x}}{1^2} = 2e^{4x}$$

Form the General Solution:

$$y = \text{C.F.} + \text{P.I.} = (A + Bx)e^{3x} + 2e^{4x}$$

Example : Solve $(D-2)^2 y = 6e^{2x}$

1. **Find the Particular Integral (**P.I.**):**

$$\text{P.I.} = \frac{1}{(D-2)^2}(6e^{2x})$$

Direct substitution $D = 2$ gives $\frac{1}{0}$, which is undefined (case of failure).

2. **Apply the Case of Failure Rule:**

Multiply by x and differentiate the denominator with respect to D:

$$\text{P.I.} = x \cdot \frac{1}{\frac{d}{dD}(D-2)^2}(6e^{2x}) = x \cdot \frac{1}{2(D-2)}(6e^{2x})$$

Substituting $D = 2$ again gives $\frac{1}{0}$. Differentiate once more and multiply by x:

$$\text{P.I.} = x^2 \cdot \frac{1}{2}(6e^{2x}) = 3x^2 e^{2x}$$

Example : Solve $(D^2+1)y = \cos 2x$

1. **Find the Complementary Function (**C.F.**):**

Auxiliary equation:

$$m^2 + 1 = 0 \Longrightarrow m = \pm i$$

$$\text{C.F.} = A\cos x + B\sin x$$

2. **Find the Particular Integral (**P.I.**):**

$$\text{P.I.} = \frac{1}{D^2+1}\cos 2x$$

Using the rule $\frac{1}{f(D^2)}\cos(ax) = \frac{1}{f(-a^2)}\cos(ax)$ where $a = 2 \Longrightarrow D^2 = -(2^2) = -4$:

$$\text{P.I.} = \frac{1}{-4+1}\cos 2x = -\frac{1}{3}\cos 2x$$

3. **Form the General Solution:**

$$y = \text{C.F.} + \text{P.I.} = A\cos x + B\sin x - \frac{1}{3}\cos 2x$$

Problem : Solve $(D^2+4)y = -10\sin(3x+2)$.

- CF $= A\cos 2x + B\sin 2x$.
- PI $= \frac{1}{D^2+4}[-10\sin(3x+2)] = \frac{-10\sin(3x+2)}{-3^2+4} = \frac{-10}{-5}\sin(3x+2) = 2\sin(3x+2)$.

Problem : Solve $(D^2+4)y = \cos 2x$ (Failure Case).

- Standard formula yields $\frac{1}{-2^2+4}\cos 2x = \frac{1}{0}$, which fails.
- **Solution:**

- Let $X = \frac{1}{D^2+4}\cos 2x$
- $Y = \frac{1}{D^2+4}\sin 2x$
- $X + iY = \frac{1}{D^2+4}e^{2ix}$
- $= \frac{1}{(D+2i)(D-2i)}e^{2ix}$
- $= \frac{1}{4i(D-2i)}e^{2ix}$
- $= \frac{1}{4i}\frac{e^{2ix}}{(D+2i-2i)} = \frac{1}{4i}e^{2ix}\frac{1}{D}\cdot 1$
- $= \frac{xe^{2ix}}{4i} = \frac{x}{4i}(\cos 2x + i\sin 2x)$
- $= \frac{x}{4}\sin 2x - i\frac{x}{4}\cos 2x$
- Now comparing real and imaginary part
- $X = \frac{x}{4}\sin 2x = y_p$

Question: Solve $x^2\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} - 4y = x^4$

Detailed Step-by-Step Solution:

Transform the Equation:

Let $x = e^z \Longrightarrow z = \ln x$. Define $D_z = \frac{d}{dz}$.

- $x\frac{dy}{dx} = D_z y$
- $x^2\frac{d^2y}{dx^2} = D_z(D_z - 1)y$

Substitute into Differential Equation:

$$D_z(D_z - 1)y - 2D_z y - 4y = (e^z)^4$$

$$(D_z^2 - D_z - 2D_z - 4)y = e^{4z}$$

$$(D_z^2 - 3D_z - 4)y = e^{4z}$$

Find Complementary Function (CF**):**

Auxiliary equation:

$$m^2 - 3m - 4 = 0 \Longrightarrow (m-4)(m+1) = 0 \Longrightarrow m_1 = 4, m_2 = -1$$

$$\text{CF} = c_1 e^{4z} + c_2 e^{-z} = c_1 x^4 + c_2 x^{-1} = c_1 x^4 + \frac{c_2}{x}$$

Find Particular Integral (PI**):**

Since $m = 4$ is a root of the auxiliary equation (case of resonance):

$$\text{PI} = \frac{1}{D_z^2 - 3D_z - 4} e^{4z} = z \cdot \frac{1}{\frac{d}{dD_z}(D_z^2 - 3D_z - 4)} e^{4z} = z \cdot \frac{1}{2D_z - 3} e^{4z}$$

Substitute $D_z = 4$:

$$\text{PI} = z \cdot \frac{e^{4z}}{2(4) - 3} = \frac{z e^{4z}}{5}$$

Substitute $z = \ln x$ and $e^{4z} = x^4$:

$$\text{PI} = \frac{1}{5} x^4 \ln x$$

General Solution:

$$y = \text{CF} + \text{PI}$$

$$y = c_1 x^4 + \frac{c_2}{x} + \frac{1}{5} x^4 \ln x$$

These relations come directly from applying the chain rule to the change of variables $x = e^z$ (or $z = \ln x$).

Here is the step-by-step mathematical proof:

1. Proving $x\frac{dy}{dx} = D_z y$

Start with the definition $z = \ln x$ and take its derivative with respect to x:

$$\frac{dz}{dx} = \frac{1}{x}$$

Now, apply the chain rule to express $\frac{dy}{dx}$ in terms of z:

$$\frac{dy}{dx} = \frac{dy}{dz} \cdot \frac{dz}{dx}$$

Substitute $\frac{dz}{dx} = \frac{1}{x}$:

$$\frac{dy}{dx} = \frac{dy}{dz} \cdot \frac{1}{x}$$

Multiply both sides by x:

$$x\frac{dy}{dx} = \frac{dy}{dz}$$

Since the operator D_z is defined as $\frac{d}{dz}$, we can write $\frac{dy}{dz}$ as $D_z y$:

$$x\frac{dy}{dx} = D_z y$$

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2. Proving $x^2\frac{d^2y}{dx^2} = D_z(D_z - 1)y$

To find the second derivative $\frac{d^2y}{dx^2}$, differentiate $\frac{dy}{dx}$ with respect to x:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dx}\left(\frac{1}{x}\frac{dy}{dz}\right)$$

Apply the product rule $\frac{d}{dx}[u \cdot v] = u'v + uv'$:

$$\frac{d^2y}{dx^2} = \left(\frac{d}{dx}\frac{1}{x}\right)\frac{dy}{dz} + \frac{1}{x}\left(\frac{d}{dx}\frac{dy}{dz}\right)$$

$$\frac{d^2y}{dx^2} = -\frac{1}{x^2}\frac{dy}{dz} + \frac{1}{x}\left(\frac{d}{dx}\frac{dy}{dz}\right)$$

Apply the chain rule again to the term $\frac{d}{dx}\left(\frac{dy}{dz}\right)$:

$$\frac{d}{dx}\left(\frac{dy}{dz}\right) = \frac{d}{dz}\left(\frac{dy}{dz}\right) \cdot \frac{dz}{dx} = \frac{d^2y}{dz^2} \cdot \frac{1}{x}$$

Substitute this back into the second derivative equation:

$$\frac{d^2y}{dx^2} = -\frac{1}{x^2}\frac{dy}{dz} + \frac{1}{x}\left(\frac{d^2y}{dz^2} \cdot \frac{1}{x}\right)$$

$$\frac{d^2y}{dx^2} = -\frac{1}{x^2}\frac{dy}{dz} + \frac{1}{x^2}\frac{d^2y}{dz^2}$$

Factor out $\frac{1}{x^2}$:

$$\frac{d^2y}{dx^2} = \frac{1}{x^2}\left(\frac{d^2y}{dz^2} - \frac{dy}{dz}\right)$$

Multiply both sides by x^2:

$$x^2\frac{d^2y}{dx^2} = \frac{d^2y}{dz^2} - \frac{dy}{dz}$$

Now, rewrite in terms of operator $D_z = \frac{d}{dz}$ where $\frac{d^2}{dz^2} = D_z^2$:

$$x^2\frac{d^2y}{dx^2} = D_z^2y - D_zy$$

Factor out D_zy:

$$x^2\frac{d^2y}{dx^2} = D_z(D_z - 1)y$$